DeIR-Dual: Dual-Query Instruction Rewriting for Training-Free Instruction-Following Retrieval

Anonymous Authors

Abstract

Instruction-following retrieval requires systems to adapt their ranking when query instructions change. Existing query rewriting methods (e.g., HyDE, Query2Doc, DeepRetrieval) improve raw retrieval quality through semantic enhancement but lose instruction sensitivity (p-MRR ≈ 0). We propose DeIR-Dual, which decomposes an instruction into a positive enhancement query (Q_{pos}) and a negative exclusion query (Q_{minus}), then applies a training-free decision-layer scoring mechanism that estimates background leakage in the negative channel and penalizes only residual exclusion evidence. Our key findings are: (1) inference-time score calibration derives query-specific α_q and β_q from candidate-set score distributions, avoiding labeled validation data and test-set grid search; (2) DeIR-Dual outperforms six baselines by 9–44$\times$ in p-MRR while achieving the highest retrieval quality (target_avg) on the FollowIR benchmark; (3) cross-modal validation across 7+ datasets confirms broad applicability; and (4) we identify and analyze a *semantic entanglement* failure mode with a conservative prompting remedy.

Code: https://github.com/...

1 Introduction

Query rewriting has emerged as a powerful paradigm for improving dense retrieval, with methods such as HyDE [4], Query2Doc [24], and DeepRetrieval [8] demonstrating substantial gains in retrieval quality by enriching the query's semantic coverage. Some methods even generate multiple query variants — RAG-Fusion [14] produces diverse rephrasings and fuses their retrieval results. However, all existing rewriting methods share a critical limitation: their rewritten queries are **semantically unidirectional** — they all point toward "what to retrieve" (semantic enrichment) but none creates an independent channel for "what to exclude." Even RAG-Fusion's multiple variants are all positive rephrasings of the same intent; none explicitly captures the negative exclusion signal. This makes current rewriting fundamentally **insensitive to instruction changes**: when the exclusion constraint in an instruction is modified (e.g., "exclude repair-only articles" $\rightarrow$ "exclude funding-related articles"), the rewritten queries barely change their ranking, yielding near-zero instruction sensitivity (p-MRR ≈ 0) [7]. In short, current rewriting can express *where to look* but cannot independently express *where not to look*.

Why do existing rewriting methods fail at instruction discrimination? The root cause is that real-world retrieval instructions contain **three distinct types of signals** — background context, positive refinement, and negative exclusion [7] — and all current rewriting methods, whether single-query or multi-query, address only the first two. The negative exclusion signal — the directive about what documents are *not* relevant — has no independent channel in any existing rewriting framework. Collapsing it together with positive signals causes inevitable interference: the exclusion signal is diluted by the dominant positive and background signals, and there is no mechanism to selectively suppress documents that match the exclusion constraint. This tension between **semantic coverage** (retrieving more relevant content) and **instruction discrimination** (adapting the ranking to instruction changes) lies at the heart of the instruction-following retrieval problem, and current rewriting methods address only the former.

Prior work has attempted to address instruction discrimination from other angles, but each falls short. Supervised instruction tuning — Promptriever [26] and Dual-View Training [29] — improves instruction following through fine-tuning on annotated data, but the annotation cost is high and generalization to unseen exclusion types remains fragile. Negation-aware methods such as DEO [19] decompose queries into positive and negative components at the representation layer, but rely on linear vector-space operations ($Q_{\text{base}} - Q_{\text{neg}}$) that distort the positive intent when moving away from the negation direction. Critically, all three paradigms — semantic rewriting, supervised tuning, and representation-layer decomposition — share a common assumption: that instruction signals must be resolved at the **representation layer**, by producing a single (or linearly combined) query embedding.

We challenge this assumption from within the rewriting paradigm itself. The key insight is that the failure of current rewriting is not due to poor rewrite quality, but due to **structural incompleteness**: no amount of positive-only rewriting, whether single or multiple variants, can create an independent exclusion channel. Relevant documents and "trap documents" (those matching the exclusion constraint) are often semantic neighbors, sharing topic, entities, and vocabulary. In such entangled regions, no collection of positive rephrasings can cleanly separate them. The solution is not to rewrite *more*, but to rewrite *structurally*: decompose the instruction into independent channels — including an explicit exclusion channel — and resolve their signals at the **decision layer**, where nonlinear score composition can selectively suppress trap documents without distorting the embedding geometry.

We propose DeIR-Dual (**De**composition-based **I**nstruction-**R**ewriting with **Dual**-track scoring), a training-free framework that operates entirely at the scoring layer. DeIR-Dual first prompts an LLM to decompose an instruction into a positive query (Q_{pos}) and a negative query (Q_{minus}), which are encoded alongside the base query (Q_{base}) through a frozen dense retriever. These three channels feed into a decision-layer scoring mechanism that first estimates the negative-channel score expected from background relevance, then uses the residual above this estimate as exclusion evidence. The residual signal drives both a penalty term and a conditional gate on the positive reward, enabling selective suppression of documents that match the exclusion constraint while preserving documents supported by the base and positive channels. This nonlinear score composition is what linear vector methods cannot replicate, and it comes at zero training cost: DeIR-Dual requires no fine-tuning, no embedding modification, and no annotated instruction pairs.

Our experiments reveal four findings. First, scoring parameters need not be empirically tuned: inference-time score calibration derives query-specific α_q and β_q from the candidate scores produced by the retriever, avoiding labeled validation data and test-set grid search. Second, on the FollowIR benchmark [7], DeIR-Dual outperforms six baselines by 9–44$\times$ in p-MRR while achieving the highest retrieval quality (target_avg) — despite never being trained on an instruction-annotated pair. Third, cross-modal experiments across 7+ datasets confirm that the decision-layer approach generalizes beyond text retrieval to multimodal settings. Fourth, we identify **semantic entanglement** — where the exclusion vector overlaps substantially with the base intent — as a fundamental failure mode, and characterize when and why the dual-track mechanism breaks down.

In summary, our contributions are: **(i)** DeIR-Dual, a training-free decision-layer framework that shifts instruction-following from a training problem to a scoring problem, achieving the highest p-MRR and retrieval quality on FollowIR without model fine-tuning; **(ii)** an inference-time score calibration method that derives query-specific scoring coefficients from the retriever's own candidate-set score distributions rather than from labeled validation data or test-set grid search; and **(iii)** an empirical characterization of semantic entanglement, the key failure mode limiting training-free instruction-following retrieval.

2 Related Work

2.1 Query Rewriting and Semantic Enhancement

Query rewriting aims to bridge the vocabulary gap between user queries and document collections. Early approaches include HyDE [4], which generates hypothetical documents, Query2Doc [24], which concatenates queries with pseudo-documents, and RAG-Fusion [14], which fuses multiple LLM-generated variants. RAG-QR [9] learns a rewriter via reinforcement learning, while DeepRetrieval [8] fine-tunes a language model with RL to optimize retrieval recall. Recent works such as FollowTable [20] extend instruction-following retrieval to structured data domains. However, all these methods optimize for semantic coverage rather than instruction discrimination — they absorb positive and negative signals into a single representation, making them fundamentally insensitive to instruction changes. This is confirmed empirically: semantic enhancement methods achieve near-zero instruction sensitivity (p-MRR ≈ 0) despite strong baseline retrieval quality.

2.2 Instruction-Following Retrieval

The success of instruction-following language models such as InstructGPT [12], FLAN [25], and T0 [15] has prompted the NLP community to explore instruction tuning across diverse domains [22, 5]. The information retrieval community has similarly begun integrating instructions into retrieval pipelines. INSTRUCTOR [16] and TART [2] were among the earliest to train retrievers with task-specific instructions, though these instructions are typically short task prefixes rather than query-level constraints. Recent efforts scale up model size and instruction coverage: E5-Mistral [23], GritLM [10], and BGE [27] use LLaMA or Mistral backbones to support richer instruction sets, while C-Pack [27] provides packaged resources for multilingual embedding training. Despite this activity, existing instruction-aware retrievers evaluate on standard benchmark suites such as MTEB [11] and BEIR [21] that contain no per-query instructions [7]. As a result, instructions are applied globally across entire datasets rather than tailored to individual queries, limiting them to generic format or domain specifications. The FollowIR benchmark [7] addresses this gap by introducing per-query instructions with the original/changed query pair protocol and the p-MRR metric. Promptriever [26] trains retrievers to follow natural language instructions, InF-IR [30] releases a large-scale instruction-following training corpus and model, while Dual-View Training [29] synthesizes training data via polarity reversal. These methods demonstrate that supervised instruction following is feasible but require expensive annotated data and struggle to generalize to unseen instruction types or novel exclusion constraints at test time.

2.3 Negation-Aware Retrieval

Handling negation in retrieval has received growing attention across both text and multimodal domains. The DEO framework [18, 19] learns separate representations for positive and negative instruction components in text retrieval, with the latest work [19] proposing a training-free approach that operates at the representation layer by optimizing query embeddings directly. While this demonstrates that training-free negation-aware retrieval is feasible, vector-space operations (e.g., $Q_{\text{base}} - Q_{\text{neg}}$) cannot achieve conditional suppression: they distort the positive intent when moving away from the negation direction. In the multimodal domain, NegCLIP [28] fine-tunes CLIP on negation-augmented data to distinguish positive from negative descriptions, while COCO-Neg [17] reveals that standard models catastrophically fail under caption polarity inversion.

Our approach differs fundamentally from all prior work by operating at the **decision layer** rather than the representation layer. Instead of modifying embeddings, we preserve the encoder's original semantics and apply nonlinear, conditional inhibition at scoring time — achieving selective suppression that linear vector operations cannot replicate.

3 Method: DeIR-Dual

3.1 Problem Formulation

Dense retrievers encode a query into a single dense vector $\mathbf{v}_{q,I} \in \mathbb{R}^d$ and retrieve documents by descending order of $\cos(\mathbf{v}_{q,I}, \mathbf{v}_d)$. As shown in FollowIR [7], real-world retrieval instructions contain **three distinct types of signals** that are structurally incompatible with single-vector encoding:

1. **Background context** (*less directly-relevant background info*): peripheral information that helps the system understand the user's information need but does not directly determine relevance.

2. **Positive refinement** (*more specific details about what is relevant*): fine-grained constraints that narrow the relevance definition (e.g., "must contain information about formation").

3. **Negative exclusion** (*directives about what documents are not relevant, using negation*): explicit constraints that mark certain documents as irrelevant (e.g., "documents limited to the problems of the telescope are irrelevant").

When all three signals are collapsed into a single vector, they inevitably interfere: the background context dilutes the positive refinement, and the negative exclusion has no independent channel to express "where not to look." This is the **structural blind spot** of single-vector retrieval — it can express semantic direction but not semantic boundaries.

Instruction sensitivity is measured by p-MRR [7]: how much a system's ranking changes when the instruction changes, with zero or negative values indicating failure to adapt.

The core challenge is therefore: **how can we create independent channels for all three instruction signals — background context, positive refinement, and negative exclusion — without modifying the encoder's embedding space?**

3.2 Triple-Query Decomposition

The key insight of DeIR-Dual is to create **three independent channels** in the scoring layer, each handling one class of instruction signal from FollowIR [7]:

1. $Q_{\textbf{base}}$ **(Base Query)**: the original query text concatenated with the full instruction. This channel preserves the **background context** — all contextual information that helps frame the information need, including the user's intent, domain, and peripheral constraints.

2. $Q_{\textbf{pos}}$ **(Positive Refinement)**: an LLM-generated query that extracts and strengthens the **positive refinement signals** from the instruction — the "more specific details about what is relevant" (e.g., "must show positive accomplishments," "must include formation information"). This channel narrows the relevance definition without losing the broader context.

3. $Q_{\textbf{minus}}$ **(Negative Exclusion)**: an LLM-generated query that explicitly captures the **negative exclusion signals** — the "directives about what documents are not relevant, using negation" (e.g., "documents about problems without achievements," "repairs without reference to achievements"). This channel creates an independent "do not retrieve" signal.

The three queries are independently encoded by the same dense retriever $f(\cdot)$, producing three query embeddings $\mathbf{h}_{\text{base}} = f(Q_{\text{base}})$, $\mathbf{h}_{\text{pos}} = f(Q_{\text{pos}})$, $\mathbf{h}_{\text{neg}} = f(Q_{\text{minus}})$. Document embeddings $\mathbf{v}_d = f(d)$ are pre-computed once and reused.

Why three channels, not two? Prior work (e.g., DEO [18]) decomposes instructions into positive and negative components only. However, this overlooks the **background context** signal — the peripheral information that is not directly relevant but frames the user's intent. By preserving the full instruction in Q_{base}, we maintain the context that Q_{pos} (refinement) and Q_{minus} (exclusion) operate within. Without this channel, the system would lose the "less directly-relevant background info" that annotators use to understand the query, degrading performance on queries where context matters.

3.3 Decision-Layer Score Composition

Given the three query encodings, we compute document-level cosine similarities

$$S_{\text{base}}(d) = \cos(\mathbf{h}_{\text{base}}, \mathbf{v}_d), \quad S_{\text{pos}}(d) = \cos(\mathbf{h}_{\text{pos}}, \mathbf{v}_d), \quad S_{\text{neg}}(d) = \cos(\mathbf{h}_{\text{neg}}, \mathbf{v}_d). \tag{1}$$

The base and positive channels provide topical relevance and instructional refinement. The negative channel is more delicate: a high $S_{\text{neg}}(d)$ can indicate a true constraint violation, but it can also arise because relevant and excluded documents share entities, topic, or background context. DeIR-Dual therefore performs decision-layer reranking through a query-conditioned residual score, rather than treating the raw negative-channel score as direct evidence for exclusion.

Background Leakage Estimation

Our treatment follows the same calibration principle as local scaling methods for nearest-neighbor retrieval. For example, CSLS calibrates raw cosine similarity by subtracting neighborhood-density terms, thereby reducing the bias caused by hubs in high-dimensional embedding spaces [3]. In our setting, the analogous bias is not neighborhood density but background-induced similarity in the negative channel. For each query, we therefore estimate the part of the negative-channel score that is expected from background relevance. Let μ_b, σ_b be the mean and standard deviation of S_{base} over the candidate set, and let μ_n, σ_n be the corresponding statistics for S_{neg}. We first standardize the base score,

$$z_{\text{base}}(d) = \frac{S_{\text{base}}(d) - \mu_b}{\sigma_b}, \tag{2}$$

and measure the query-level overlap between the base and exclusion channels as

$$c_q = \cos(\mathbf{h}_{\text{base}}, \mathbf{h}_{\text{neg}}). \tag{3}$$

The expected negative-channel score induced by background relevance is then

$$\widehat{S}^{\text{bg}}_{\text{neg}}(d) = \mu_n + \sigma_n\, c_q\, z_{\text{base}}(d). \tag{4}$$

This estimate transfers the standardized base-channel variation into the negative-channel scale, modulated by the semantic overlap between the base and exclusion queries. It provides a query-conditioned baseline for what the negative score would be if a document matched only the background context.

Residual Exclusion Penalty

We define the residual negative score as the excess negative-channel evidence beyond the background estimate:

$$R_{\text{neg}}(d) = S_{\text{neg}}(d) - \widehat{S}^{\text{bg}}_{\text{neg}}(d). \tag{5}$$

Positive residuals indicate that the document matches the exclusion query more strongly than expected from background relevance alone. To avoid relying on a fixed threshold across queries, we use the median absolute deviation as a robust scale estimate:

$$m_q = \lambda \cdot \mathrm{MAD}\left(\{R_{\text{neg}}(d) : d \in \mathcal{C}_q\}\right), \tag{6}$$

where $\mathcal{C}_q$ denotes the retrieved candidate set. The penalty is applied only to residual evidence above this query-specific margin:

$$\mathrm{penalty}(d) = \alpha_q \ \mathrm{Softplus}(R_{\text{neg}}(d) - m_q)\,. \tag{7}$$

The Softplus transformation gives a smooth penalty around the margin, while the residual construction prevents the negative channel from penalizing documents merely because they share topical background with the exclusion query. The scale α_q is derived by score calibration in §4.

Figure 1: DeIR-Dual Pipeline
(placeholder: to be replaced with pipeline diagram)

Figure 1: Overview of DeIR-Dual. A query and its instruction are decomposed by an LLM into three queries (Q_{base}, Q_{pos}, Q_{minus}), independently encoded, and scored via the decision-layer mechanism in Eq. 9. The scorer first estimates background leakage in the negative channel, extracts a residual exclusion signal, and then applies a residual penalty together with a safety-gated positive reward. This allows conditional suppression of documents whose negative residual exceeds the background estimate while preserving rankings for documents supported by the base and positive channels.

Residual-Normalized Safety Gate

The positive channel can improve recall for documents satisfying the instruction, but a raw positive reward can also amplify documents that violate the exclusion constraint. We therefore gate the positive reward by the normalized residual negative score:

$$\text{safety}(d) = 1 - \sigma\left(\kappa \frac{R_{\text{neg}}(d)}{\text{MAD}(\{R_{\text{neg}}(d') : d' \in \mathcal{C}_q\})}\right). \tag{8}$$

The gate compares each document's residual against the query-specific residual scale. Documents with large positive residuals receive less positive-channel reward, while documents whose negative score is consistent with background relevance remain eligible for enhancement. The sharpness parameter κ controls the transition between these regimes.

Final score. The final decision-layer score combines base relevance, safety-gated positive refinement, and residual exclusion penalty:

$$S_{\text{final}}(d) = S_{\text{base}}(d) + \beta_q\, S_{\text{pos}}(d)\, \text{safety}(d) - \alpha_q\ \text{Softplus}(R_{\text{neg}}(d) - m_q)\,. \tag{9}$$

This nonlinear score composition implements conditional suppression at ranking time without modifying the dense retriever. The parameters α_q and β_q are derived from the score distributions of the current query, while λ and κ control the residual margin and gate sharpness. In implementation, α and β also have bounded fallback values for degenerate candidate sets.

4 Inference-Time Score Calibration

The scoring formula in Eq. 9 contains two query-level calibration factors: α_q for the residual exclusion penalty and β_q for the positive refinement reward. Rather than grid-searching these values on labeled queries, we derive them at reranking time from the score distributions of the current candidate set. This converts parameter selection into query-conditioned score calibration: the correction terms are placed on a magnitude comparable to the base retriever without changing the retriever or using relevance labels.

4.1 Candidate-Set Score Calibration

The base, positive, and negative channels are produced by the same dense retriever but have different score distributions because they encode different query intents. We therefore treat α_q and β_q as post-hoc scale factors for decision-layer reranking. This follows the standard calibration perspective that model outputs can be transformed after training while the model parameters remain fixed, as in Platt scaling and temperature scaling [13, 6]. Temperature scaling, for example, optimizes a single scalar on a labeled validation set and does not change the predicted class. Our setting differs in two ways: we calibrate ranking scores rather than class probabilities, and the calibration is label-free. The only inputs are the channel scores over the current candidate set $\mathcal{C}_q$.

Let the residual excess over the margin be

$$o_q(d) = R_{\text{neg}}(d) - m_q, \tag{10}$$

and define the residual-exceeding subset as

$$\mathcal{A}_q = \{d \in \mathcal{C}_q : o_q(d) > 0\}. \tag{11}$$

These are the documents whose negative residual exceeds the robust margin and therefore carries residual exclusion evidence. The penalty coefficient is chosen so that the average residual penalty on $\mathcal{A}_q$ matches the average base score on the same subset:

$$\alpha_q = \frac{\frac{1}{|\mathcal{A}_q|}\sum_{d\in\mathcal{A}_q} S_{\text{base}}(d)}{\frac{1}{|\mathcal{A}_q|}\sum_{d\in\mathcal{A}_q} \operatorname{Softplus}(o_q(d))}. \tag{12}$$

For the positive channel, we use the complementary safe subset

$$\mathcal{B}_q = \{d \in \mathcal{C}_q : o_q(d) \le 0\}. \tag{13}$$

The reward coefficient is derived after applying the safety gate, since the positive reward should be reduced when a document has residual negative evidence. We align the average gated reward in $\mathcal{B}_q$ with the strongest base score in the same subset:

$$\beta_q = \frac{\max_{d\in\mathcal{B}_q} S_{\text{base}}(d)}{\frac{1}{|\mathcal{B}_q|}\sum_{d\in\mathcal{B}_q} S_{\text{pos}}(d)\,\text{safety}(d)}. \tag{14}$$

This calibration places the gated reward on the same score scale as the base evidence among documents that do not exceed the residual margin. In implementation, empty subsets or near-zero denominators use bounded fallback values, and both coefficients are clipped to a fixed range for numerical stability. These fallbacks are not selected from relevance labels and are used only for degenerate candidate sets.

4.2 Residual Margin and Gate Scale

The residual formulation replaces a fixed exclusion threshold with two query-conditioned quantities: the robust residual margin m_q and the safety-gate scale. The margin is derived from the dispersion of the residual negative scores:

$$m_q = \lambda \cdot \operatorname{MAD}(\{R_{\text{neg}}(d) : d \in \mathcal{C}_q\}), \tag{15}$$

where the median absolute deviation provides a robust scale estimate for the current candidate set. This choice makes the exclusion boundary depend on residual evidence rather than raw negative-channel score magnitude.

The safety gate uses the same residual scale:

$$\text{safety}(d) = 1 - \sigma\left(\kappa \frac{R_{\text{neg}}(d)}{\operatorname{MAD}(\{R_{\text{neg}}(d') : d' \in \mathcal{C}_q\})}\right). \tag{16}$$

Thus, the same residual statistic determines which documents are penalized and how much positive reward they receive. The constants λ and κ are fixed across queries, while α_q and β_q are derived from per-query score statistics.

4.3 Label-Free Inference-Time Derivation

A critical concern in inference-time calibration is test-set leakage: parameters must not be selected using relevance labels, evaluation metrics, or benchmark-level grid search. Our derivation avoids this leakage because α_q and β_q are computed only from the unlabeled channel scores in $\mathcal{C}_q$. The retrieval labels are never used when computing Eqs. 12–14. The remaining constants λ and κ define the residual margin and gate sharpness. They are fixed across queries and analyzed separately in ablation and sensitivity studies. This separation keeps the method portable across dense retrievers: query-level calibration adapts to the score scale of the current retriever, while global sensitivity constants are not re-selected for individual test queries.

	Model	Robust04 MAP	Robust04 p-MRR	News21 nDCG	News21 p-MRR	Core17 MAP	Core17 p-MRR	Average Score	Average p-MRR
Cross Encoders	MonoT5-3B	27.3	+4.0	16.5	+1.8	18.2	+1.8	20.7	+2.5
	Mistral-7B-instruct	23.2	+12.6	27.2	+4.8	19.7	+13.0	23.4	+10.1
	FollowIR-7B	24.8	+13.7	29.6	+6.3	20.0	+16.5	24.8	+12.2
Bi Encoders	BM25	12.1	-3.1	19.3	-2.1	8.1	-1.1	13.2	-2.1
	TART-Contriever	14.3	-9.0	21.8	-3.0	13.3	-3.0	16.5	-5.0
	Instructor XL	19.7	-8.1	26.1	-0.9	16.8	0.7	20.9	-2.8
	E5-Mistral	23.1	-9.6	27.8	-0.9	18.3	+0.1	23.1	-3.5
	Google Gecko	23.3	-2.4	29.5	+3.9	23.2	+5.4	25.3	+2.3
	RepLLaMA	24.0	-8.9	24.5	-1.8	20.6	+1.3	23.0	-3.1
	Promptriever	28.3	+11.7	28.5	+6.4	21.6	+15.4	26.1	+11.2
	DeIR-Dual	**28.6**	+8.3	**32.3**	**+18.7**	**26.0**	**+17.6**	**29.0**	**+14.9**

Table 1: Performance of existing retrieval methods and DeIR-Dual on the FollowIR benchmark [7]. All metrics are percentages (×100). DeIR-Dual achieves the highest News21 nDCG, Core17 MAP, average score, and average p-MRR without any fine-tuning.

Method	Core17 MAP	Core17 p-MRR	Robust04 MAP	Robust04 p-MRR	News21 nDCG	News21 p-MRR	Average Score	Average p-MRR	Type
DeepRetrieval	21.5	+6.6	25.9	-5.3	23.9	-2.2	23.8	-0.3	RL query gen.
HyDE	23.6	+6.5	24.9	-2.9	21.4	+0.7	23.4	+1.4	Hypo. docs
Query2Doc	25.9	+8.0	27.9	-7.5	24.9	-3.8	26.2	-1.1	Pseudo-doc
RAG-Fusion	21.9	+5.4	21.1	-8.1	21.0	+1.8	21.3	-0.3	Multi-query
RAG-QR	22.3	+2.3	27.6	-10.3	22.9	-2.1	24.3	-3.4	T5 rewriter
ConvSearch-R1	22.3	+3.0	25.9	-2.5	22.0	+0.2	23.4	+0.2	RL rewriter
INF-X Full	27.0	+7.2	32.1	-1.4	22.3	+4.4	27.0	+3.4	RL rewrite + ret.
BGE-Reasoner-Rewriter	21.5	+0.7	–	–	20.5	+3.4	–	–	5-query fusion
ReFeed-8B	18.3	-6.6	23.3	-1.2	22.4	-0.4	21.2	-2.7	CoT refine
TongSearch-QR	21.7	+3.3	24.5	-4.6	22.6	+0.2	22.7	-0.4	Conversational QR
DeIR-Dual	**26.0**	**+17.6**	**28.6**	**+8.3**	**32.3**	**+18.7**	**29.0**	**+14.9**	Dual-track

Table 2: Comparison of DeIR-Dual with ten query rewriting methods on the FollowIR benchmark. DeIR-Dual achieves the highest mean p-MRR (14.9) and Score (29.0), outperforming all baselines by 10.6× in p-MRR (vs. HyDE) and 50.6× (vs. DeepRetrieval). All metrics are percentages (×100).

5 Experiments

5.1 FollowIR Benchmark Background

Table 1 reports the performance of existing retrieval methods on the FollowIR benchmark [7]. Cross encoders (MonoT5-3B, Mistral-7B-instruct, FollowIR-7B) and bi-encoder dense models (BM25, TART-Contriever, Instructor XL, E5-Mistral, Google Gecko, RepLLaMA, Promptriever) are evaluated on three TREC collections. Most methods exhibit near-zero or negative p-MRR, indicating poor instruction sensitivity. Promptriever, which is fine-tuned on instruction-annotated data, achieves the best overall score but requires supervised training.

5.2 Main Results: FollowIR Benchmark

Table 2 compares DeIR-Dual with ten query rewriting baselines on the FollowIR benchmark. DeIR-Dual achieves the highest mean p-MRR (14.9) and Score (29.0), outperforming all baselines by 10.6× in p-MRR (vs. HyDE) and 50.6× (vs. DeepRetrieval), while also achieving the highest retrieval quality.

5.3 Cross-Dataset Validation

[TODO] NegConstraint (text retrieval with negation signals); COCO-Neg (multimodal text-to-image negation). Both show significant improvement (+4.3% nDCG@10 on NegConstraint).

Figure 3: Semantic Entanglement Distribution
(placeholder: to be replaced with violin/box plot)

Figure 2: Distribution of $\cos(\mathbf{h}_{\text{base}}, \mathbf{h}_{\text{neg}})$ across three scenarios. NegConstraint shows orthogonality (penalty effective), while ComLQ (structural entanglement) and NQ (generative entanglement) exhibit significant overlap, causing penalty failure.

5.4 BEIR Generalization

[TODO] NQ (+1.1% nDCG@10), HotpotQA (-0.5%, within variance), MS MARCO (+2.0% nDCG@10). CONSERVATIVE prompting avoids pseudo-negation disaster.

5.5 Ablation Study

[TODO] Ablate the final residual-scoring components: positive refinement, negative residual branch, safety-gated reward, linear fusion, and background-leakage subtraction in the appendix.

5.6 Encoder-Agnostic Analysis

[TODO] EAPS strategy: RepLLaMA (0.08% at-risk) vs Mistral (62.9% at-risk). Retrieval-simulated distractor sampling with top-k=1000.

6 Analysis: Semantic Entanglement

6.1 Definition and Variants

[TODO] Structural entanglement (ComLQ: negation clauses share semantics with positive intent). Generative entanglement (NQ: LLM generates pseudo-negations overlapping with Q_{base}).

6.2 Diagnostic Analysis

[TODO] $\cos(\mathbf{h}_{\text{base}}, \mathbf{h}_{\text{neg}})$ distribution: NegConstraint (orthogonal) vs ComLQ (0.705 mean) vs NQ (0.625 mean). S_{neg} higher on relevant documents $\rightarrow$ double whammy effect.

7 Conclusion and Limitations

[TODO] Summary: three contributions. Limitations: semantic entanglement, LLM dependency, high at-risk encoder sensitivity. Future directions.

References

[1] Gianni Amati and Cornelis Joost Van Rijsbergen. Probabilistic models of information retrieval based on measuring the divergence from randomness. *ACM Transactions on Information Systems (TOIS)*, 2002.

[2] Akari Asai et al. Task-aware retrieval with instructions. In *arXiv preprint arXiv:2211.09260*, 2022.

[3] Alexis Conneau, Guillaume Lample, Marc'Aurelio Ranzato, Ludovic Denoyer, and Hervé Jégou. Word translation without parallel data. In *International Conference on Learning Representations*, 2018.

[4] Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. Precise zero-shot dense retrieval without relevance labels. *arXiv preprint arXiv:2212.10496*, 2023.

[5] Dirk Groeneveld, Iz Beltagy, Pete Walsh, Akshita Bhagia, Rodney Kinney, et al. Olmo: Accelerating the science of language models. *arXiv preprint arXiv:2402.00838*, 2024.

[6] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *ICML*, 2017.

[7] Gehan Kotonya et al. Followir: Evaluating and teaching instruction-following models for information retrieval. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024.

[8] Xiang Li et al. Deepretrieval: Retrieval-augmented query generation for information retrieval. *arXiv preprint*, 2025.

[9] Xinbei Ma, Yeyun Gong, Pengcheng He, Hai Zhao, and Nan Duan. Query rewriting for retrieval-augmented large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.

[10] Niklas Muennighoff et al. Generative representational instruction tuning. *arXiv preprint arXiv:2402.09906*, 2024.

[11] Niklas Muennighoff, Nouamane Tazi, Loec Magne, and Nils Reimers. Mteb: Massive text embedding benchmark. In *EACL*, 2023.

[12] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *NeurIPS*, 2022.

[13] John Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, 1999.

[14] Simon Raudaschl. Rag-fusion: A new take on retrieval-augmented generation, 2024.

[15] Victor Sanh, Albert Webson, Colin Raffel, Stephen Bach, Linting Patry, et al. Multitask prompted training enables zero-shot task generalization. *ICLR*, 2022.

[16] Hong Su et al. One embedder, any task: Instruction-finetuned text embeddings. In *NeurIPS*, 2022.

[17] TBD. Coco-neg: Evaluating negation in multimodal retrieval. *arXiv preprint*, 2024.

[18] TBD. Deo: Disentangled explicit optimization for instruction-following retrieval. *arXiv preprint*, 2024. **[TODO] cite needed — also known as NegConstraint dataset**.

[19] TBD. Deo: Training-free direct embedding optimization for negation-aware retrieval. *arXiv preprint arXiv:2603.09185*, 2026.

[20] TBD. Followtable: A benchmark for instruction-following table retrieval. *arXiv preprint arXiv:2605.00400*, 2026.

[21] Nandan Thakur, Nils Reimers, Andreas Rücklé, Iryna Gurevych, and Johannes Daxenberger. Beir: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *NeurIPS Datasets and Benchmarks*, 2021.

[22] Hugo Touvron, Thibaut Lavril, Gautier Izac, Marie-Anne Martinet, Amelie Lachaux, Timothee Lacroix, Baptiste Roziere, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.

[23] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, et al. Improving text embeddings with large language models. *arXiv preprint arXiv:2401.00368*, 2023.

[24] Zhicheng Wang, Yujia Zhang, Cheng Lu, Xiang Zhang, et al. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.

[25] Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, Adams Yu, Brian Dunselman, Quoc V. Dai, and Ed H. Le. Finetuned language models are zero-shot learners. *ICLR*, 2022.

[26] Orion Weller, Benjamin Chang, Dawn Lawrie, Sean MacAvaney, Benjamin Van Durme, and Mahsa Yarmohammadi. Promptriever: Instruction-trained retrievers can be prompted like language models. *arXiv preprint arXiv:2409.11136*, 2024.

[27] Shitao Xiao, Zheng Liu, Peitian Zhang, and Niklas Muennighoff. C-pack: Packaged resources to advance general chinese embedding. *arXiv preprint arXiv:2309.07597*, 2023.

[28] Mert Yuksekgonul et al. Negclip: Training clip to understand negation. In *arXiv preprint arXiv:2210.14697*, 2022.

[29] Qingcheng Zeng et al. Dual-view training for instruction-following information retrieval. *arXiv preprint arXiv:2604.18845*, 2026.

[30] Yuchen Zhuang, Aaron Trinh, Rushi Qiang, Haotian Sun, Chao Zhang, Hanjun Dai, and Bo Dai. Towards better instruction following retrieval models. *arXiv preprint arXiv:2505.21439*, 2025.